



AVERAGE

Concept Clarity Sheet

Quantitative Aptitude & Reasoning • All Formulas + Solved Examples

1. What is an Average?

Average (also called the Arithmetic Mean) is a single value that represents the central value of a group of numbers. It tells us what value each item would have if the total were shared equally among all items.

$$\text{Average} = \text{Sum of all Observations} \div \text{Number of Observations}$$

Example: Marks obtained by a student in 5 subjects are 60, 70, 80, 90, 100. Find the average.

$$\text{Sum} = 60 + 70 + 80 + 90 + 100 = 400$$

$$\text{Number of observations} = 5$$

$$\text{Average} = 400 \div 5 = 80$$

2. Two Formulas Every Student Must Know

From the basic definition, two very useful formulas can be derived. Most average questions are solved by simply rearranging these.

$$\text{Sum of Observations} = \text{Average} \times \text{Number of Observations}$$

$$\text{Number of Observations} = \text{Sum of Observations} \div \text{Average}$$

Example: The average weight of 25 students is 48 kg. Find the total weight of all students.

$$\text{Total weight} = \text{Average} \times \text{Number of students}$$

$$\text{Total weight} = 48 \times 25 = 1200 \text{ kg}$$

3. Average of Consecutive Numbers / Natural Numbers

(a) Average of n consecutive numbers

For a set of consecutive numbers (or numbers in Arithmetic Progression), the average is simply the middle value, or the average of the first and last term.

$$\text{Average} = (\text{First term} + \text{Last term}) \div 2$$

(b) Average of first n natural numbers



$$\text{Average} = (n + 1) \div 2$$

(c) Average of first n even numbers

$$\text{Average} = (n + 1)$$

(d) Average of first n odd numbers

$$\text{Average} = n$$

(e) Average of squares of first n natural numbers

$$\text{Average} = (n + 1)(2n + 1) \div 6$$

Example: Find the average of the first 20 natural numbers.

$$\text{Average} = (n + 1) \div 2 = (20 + 1) \div 2 = 21 \div 2 = 10.5$$

Example: Find the average of the first 15 odd numbers.

$$\text{Average of first } n \text{ odd numbers} = n$$

$$\text{Average} = 15$$

4. Weighted Average

Used when different groups (with different sizes) are combined and each group has its own average. A simple average cannot be used here — each group's average must be “weighted” by its size.

$$\text{Weighted Average} = (n_1A_1 + n_2A_2 + \dots + n_kA_k) \div (n_1 + n_2 + \dots + n_k)$$

where n = number of items in each group, A = average of that group

Example: A class has 30 boys with an average weight of 45 kg and 20 girls with an average weight of 40 kg. Find the average weight of the whole class.

$$\text{Weighted Average} = (30 \times 45 + 20 \times 40) \div (30 + 20)$$

$$= (1350 + 800) \div 50$$

$$= 2150 \div 50 = 43 \text{ kg}$$

5. Combined Average of Two Groups — Shortcut Method

This alligation-style shortcut avoids long calculation when two groups are combined.

$$\text{Combined Average} = A_2 + [n_1 \div (n_1 + n_2)] \times (A_1 - A_2)$$



where A_1, A_2 are the two group averages and n_1, n_2 are their sizes

Note: This is simply a rearranged form of the weighted-average formula and gives the same answer — useful for quick mental calculation in exams.

6. Average Speed (Equal Distance Case)

A very frequently tested concept. Average speed is NOT the simple average of the speeds — it must be calculated using total distance and total time.

$$\text{Average Speed} = \text{Total Distance} \div \text{Total Time}$$

Special case — same distance covered at two different speeds

$$\text{Average Speed} = (2 \times S_1 \times S_2) \div (S_1 + S_2)$$

(This is the Harmonic Mean of the two speeds — used ONLY when distance is equal, not time.)

Example: A car travels a certain distance at 40 km/h and returns the same distance at 60 km/h. Find its average speed for the whole journey.

Since distance is equal on both legs, use: $\text{Average Speed} = (2 \times S_1 \times S_2) \div (S_1 + S_2)$

$$= (2 \times 40 \times 60) \div (40 + 60)$$

$$= 4800 \div 100 = 48 \text{ km/h}$$

Common mistake: $(40 + 60) \div 2 = 50 \text{ km/h}$ is WRONG — that formula only applies for equal time, not equal distance.

7. Effect of Replacing, Including or Excluding an Observation

(a) One observation is replaced by another

$$\text{New Value} = \text{Old Value} \pm (\text{Number of Observations} \times \text{Change in Average})$$

(Use + if average increases, – if average decreases)

Example: The average age of 10 students increases by 2 years when a student aged 18 is replaced by a new student. Find the age of the new student.

$$\text{Increase in total age} = 10 \times 2 = 20$$

$$\text{Age of new student} = 18 + 20 = 38 \text{ years}$$

(b) A new observation is included in the group

$$\text{New Member's Value} = \text{New Average} + (\text{Number of Old Members} \times \text{Change in Average})$$

(Use + if average increases, – if average decreases, after inclusion)



Example: The average weight of 24 students in a class is 35 kg. When the teacher's weight is included, the average increases by 1 kg. Find the teacher's weight.

Number of old members = 24, Change in average = 1, New average = $35 + 1 = 36$

Teacher's weight = New average + (Old members $\times$ Change in average)

= $36 + (24 \times 1) = 36 + 24 = 60$ kg

(c) An observation is excluded (removed) from the group

Excluded Value = Old Average – (Number of Remaining Members $\times$ Change in Average)
(Use – if average increases after removal, + if average decreases after removal)

Example: The average age of a family of 6 members is 22 years. If the age of the youngest member is excluded, the average age of the remaining 5 members becomes 25 years. Find the age of the youngest member.

Total age of 6 members = $22 \times 6 = 132$

Total age of remaining 5 members = $25 \times 5 = 125$

Age of youngest member = $132 - 125 = 7$ years

8. Average Problems Based on Age

These questions combine the basic average formula with the idea that after 't' years, every person's age increases by 't' years — so the average also increases by 't'.

Average age after t years = (Current Average Age) + t

Example: The average age of a family of 5 members is 24 years. After 4 years, a child is born. If the average age of the family is still 24 years, find the age of the child.

Total age of 5 members after 4 years = $(24 \times 5) + (5 \times 4) = 120 + 20 = 140$

Total members now = 6, Required average = 24

Total age needed for 6 members = $24 \times 6 = 144$

Age of child = $144 - 140 = 4$ years

9. Quick Tips & Shortcuts for Exams

- Average always lies between the smallest and the largest value in the data set — use this to eliminate wrong options quickly.
- If all values in a set are increased/decreased by the same number 'k', the average also increases/decreases by 'k'.
- If all values are multiplied or divided by the same number 'k', the average is also multiplied or divided by 'k'.
- For numbers in Arithmetic Progression (equally spaced), average = middle term (if odd count) or average of the two middle terms (if even count).
- Average speed uses Total Distance $\div$ Total Time — never simply average the speeds unless time is equal, not distance.



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- 'Increase in total' = Number of items $\times$ Increase in average — this single idea solves most replacement/inclusion/exclusion questions.

10. Practice Questions (Assignment)

- The average of 7 consecutive numbers is 20. Find the largest number.
- The average marks of 40 students in a class is 60. If the average marks of boys is 65 and that of girls is 55, and there are equal numbers of boys and girls, verify the class average.
- A man covers a distance at 30 km/h and returns at 20 km/h covering the same distance. Find his average speed for the whole journey.
- The average of 5 numbers is 18. If one number is excluded, the average becomes 16. Find the excluded number.
- The average temperature of a city for the first 4 days of a week was 38°C and for the last 4 days was 40°C . If the average for the whole week was 39°C , find the temperature on the 4th day.

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